

NIAMEY, Niger

WMO: 610520

Lat: 13.482N

Lon: 2.184E

Elev: 223

StdP: 98.67

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.0	17.0	-12.8	1.3	32.5	-11.0	1.5	31.4	10.3	24.6	9.4	25.0	2.5	0	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.8	42.8	20.3	41.9	20.2	41.0	20.3	27.0	32.9	26.5	32.5	26.1	32.2	3.4	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.8	29.2	25.1	20.8	28.9	24.8	20.4	28.7	86.5	33.3	84.2	32.7	82.2	32.1	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.8	6.9	13.5	44.3	1.5	0.6	12.5	44.8	11.6	45.1	10.8	45.5	9.7	45.9		
(5)			5.8	28.0	0.6	1.2	5.4	28.8	5.0	29.5	4.7	30.2	4.2	31.0	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.1	24.9	28.0	31.9	34.9	34.8	32.4	29.7	28.3	29.8	31.6	29.1	25.8	(6)	(7)
	DBStd	3.68	2.20	2.77	2.29	1.75	2.01	2.30	2.11	1.79	2.04	1.57	1.88	2.01	(7)	(8)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	(9)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	(10)
	CDD10.0	7341	463	504	680	748	768	672	611	567	593	671	574	491	(10)	(11)
	CDD18.3	4299	204	271	422	498	509	422	353	309	343	412	324	233	(11)	(12)
	CDH23.3	61132	2259	3533	6497	8333	8426	6411	4609	3548	4439	6013	4406	2659	(12)	(13)
	CDH26.7	37624	1130	2079	4291	5963	5964	4102	2419	1559	2321	3689	2692	1415	(13)	(14)
(14) Wind	WSAvg	3.3	3.6	3.5	3.3	3.1	3.8	4.0	3.7	3.0	2.8	2.5	2.5	3.3	(14)	(15)
(15) Precipitation	PrecAvg	539	0	0	3	7	30	73	147	176	88	16	0	0	(15)	(16)
	PrecMax	874	0	1	43	56	109	206	330	328	240	99	6	5	(16)	(17)
	PrecMin	292	0	0	0	0	0	2	40	14	24	0	0	0	(17)	(18)
	PrecStd	140	0	0	8	14	29	41	55	72	47	22	1	1	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	40.9	42.3	43.9	43.8	41.9	38.8	36.2	38.4	40.1	39.2	37.2	(19)	(20)
		MCWB	16.1	17.2	17.9	20.0	20.9	22.6	23.5	24.6	24.0	20.9	18.5	17.6	(20)	(21)
	2%	DB	35.2	39.1	41.5	42.9	42.5	40.2	37.0	34.9	37.1	39.1	38.3	36.0	(21)	(22)
		MCWB	15.5	17.0	17.7	20.0	21.7	23.2	24.1	24.9	24.4	21.1	18.2	16.8	(22)	(23)
	5%	DB	34.0	37.8	40.8	42.1	41.7	39.1	35.8	33.8	36.0	38.2	37.8	34.9	(23)	(24)
		MCWB	15.0	16.6	17.5	19.7	22.1	23.5	24.2	24.9	24.7	21.1	18.0	16.1	(24)	(25)
	10%	DB	32.6	36.0	39.6	41.2	40.5	37.9	34.6	32.7	34.8	37.3	36.8	33.5	(25)	(26)
		MCWB	14.6	15.9	17.2	19.8	22.6	23.6	24.2	24.8	24.8	21.8	17.7	15.3	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.3	21.1	23.6	26.3	27.5	27.1	27.2	27.4	27.7	26.9	23.3	19.3	(27)	(28)
		MCDB	33.3	32.3	32.5	36.7	36.4	34.5	32.7	31.7	32.9	32.9	33.8	35.3	(28)	(29)
	2%	WB	16.7	18.9	21.8	25.2	26.4	26.2	26.4	26.5	26.9	26.1	21.7	17.9	(29)	(30)
		MCDB	33.0	34.8	34.2	36.0	35.7	33.8	31.8	31.0	32.0	32.4	33.2	33.4	(30)	(31)
	5%	WB	15.7	17.4	19.9	24.4	25.6	25.7	25.9	26.0	26.3	25.4	20.3	16.8	(31)	(32)
		MCDB	32.3	35.7	36.4	35.5	35.5	33.5	31.2	30.5	31.7	32.2	33.7	32.7	(32)	(33)
	10%	WB	14.8	16.3	18.2	23.6	25.0	25.2	25.4	25.6	25.8	24.8	18.9	15.9	(33)	(34)
		MCDB	31.1	35.0	37.5	35.0	35.1	33.2	30.8	29.9	31.3	32.0	32.9	31.5	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	14.8	14.4	12.8	11.1	10.4	9.1	8.3	9.5	12.0	15.3	15.0	(35)	(36)
		MCDBR	16.6	16.4	15.7	13.7	12.3	11.4	10.4	9.5	10.9	13.0	15.6	16.3	(36)	(37)
	5% WB	MCWBR	6.6	6.1	5.9	6.2	5.4	3.7	3.1	3.1	3.5	4.8	5.2	6.1	(37)	(38)
		MCWBR	15.6	15.0	13.2	11.3	10.9	10.3	8.9	8.5	9.6	10.8	14.0	15.8	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.601	0.648	0.730	0.739	0.721	0.690	0.623	0.590	0.580	0.627	0.552	0.542	(39)	(40)
		taud	1.692	1.563	1.429	1.500	1.535	1.606	1.787	1.850	1.860	1.714	1.765	1.785	(40)	(41)
	Ebn at Noon		705	695	652	647	647	658	708	741	751	701	740	741	(41)	(42)
		Edh at Noon	236	279	325	302	287	263	221	210	208	236	217	210	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		5.61	6.11	6.43	6.52	6.38	6.06	5.81	5.60	6.04	6.04	5.77	5.46	(43)	(44)
	RadStd		0.22	0.26	0.32	0.26	0.18	0.24	0.16	0.22	0.20	0.18	0.13	0.17	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	+0.37	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page